Project Report: Synthetic Dataset Criteria & Verified Source Mapping

Agentic AI for Pregnancy Monitoring: Preeclampsia Risk Triage & Macrosomia Risk Prevention

Prepared for Dataset Design, Model Training, and Retraining Strategy

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Abstract

This report defines the final dataset schema (24 columns) and documents synthetic data generation criteria for a pregnancy monitoring project targeting (i) preeclampsia risk detection and triage and (ii) macrosomia risk reduction through glycemic management. Since real patient datasets are limited due to privacy and access constraints, a clinically grounded synthetic dataset is proposed. For every dataset column, this report provides: the range/format used to generate synthetic values, column frequency (live vs periodic vs profile), and the verified clinical source(s) used to justify the criteria. Only authoritative sources such as ACOG and NCBI StatPearls are used.

1 Introduction

Pregnancy complications such as **preeclampsia** can rapidly escalate and require urgent medical triage. Additionally, **gestational diabetes and maternal hyperglycemia** contribute to increased fetal growth and macrosomia risk. This project uses structured live patient monitoring data to compute:

- **Preeclampsia risk score** → mapped to Low/Medium/High,
- **Macrosomia risk score** → mapped to Low/Medium/High.

The dataset is designed for:

- frequent **live monitoring** records,
- periodic **clinical/lab information**,
- straightforward **model retraining** as new records accumulate.

2 Verified Medical Sources (Direct Links)

Table 1 lists the verified sources used. These are clinical-grade references from recognized medical organizations.

Table 1: Verified Sources Used for Criteria Extraction

Source	Link
ACOG Practice Bulletin 222	https://www.preeclampsia.org/frontend/assets/img/advocacy_resource/Gestational_Hypertension_and_Preeclampsia_ACOG_Practice_Bulletin%2C_Number_222_1605448006.pdf
ACOG Diabetes FAQ	https://www.acog.org/womens-health/faqs/pregnancy-with-type-1-or-type-2-diabetes
NCBI StatPearls: Preeclampsia	https://www.ncbi.nlm.nih.gov/books/NBK570611/
NCBI StatPearls: Hypertension in Pregnancy	https://www.ncbi.nlm.nih.gov/books/NBK430839/
NCBI StatPearls: Gestational Diabetes	https://www.ncbi.nlm.nih.gov/books/NBK545196/
NCBI StatPearls: Macrosomia	https://www.ncbi.nlm.nih.gov/books/NBK557577/

3 Final Dataset Schema (24 Columns)

The dataset contains:

- **20 input columns** (live + periodic + profile),
- **2 categorical tier labels** (preeclampsia and macrosomia),
- **2 optional numeric score labels** for regression calibration.

Table 2: Final Dataset Column Count Summary

Category	Count
Input features	20
Tier labels (classification targets)	2
Optional numeric score labels	2
Total Columns	**24**

4 Frequency Grouping: Live vs Periodic vs Profile

The dataset is optimized for live patient data. Periodic columns are expected to be missing in many rows.

Table 3: Column Frequency Classification

Frequency Type	Columns
Live (Frequent)	timestamp, systolic_bp, diastolic_bp, severe_headache, vision_changes, upper_abdominal_pain, swelling_face_hands, fasting_glucose, postprandial_1hr_glucose, food_log_text, post_meal_walk_minutes, sleep_hours
Periodic (Occasional)	proteinuria_dipstick, hba1c, weight_kg
Profile (Mostly constant per patient)	patient_id, gestational_age_weeks, maternal_age, pre_pregnancy_bmi, gdm_diagnosed
Targets / Labels	preeclampsia risk tier, macrosomia risk tier, preeclampsia risk score label, macrosomia risk score label

5 Column-wise Synthetic Criteria Mapping (24 Columns)

This is the main mapping sheet: criteria + verified source + extracted fact.

Table 4: Synthetic Criteria for Each Column with Verified Source and Extracted Fact

Column	Frequency	Criteria Used	Verified Source Link + Extracted Fact Used
patient id	Profile	P0001–P0500	System-generated ID (no clinical criteria required).
timestamp	Live	Multiple records/day	System-generated time-series tracking.
gestational age weeks	Profile	20–40 (main)	StatPearls PE: evaluated after ≥20 weeks. https://www.ncbi.nlm.nih.gov/books/NBK570611/
systolic bp	Live	90–180; HTN ≥140; Severe ≥160	ACOG PB 222: severe SBP ≥160. https://www.preeclampsia.org/frontend/assets/img/advocacy_resource/Gestational_Hypertension_and_Preeclampsia_ACOG_Practice_Bulletin%2C_Number_222_1605448006.pdf
diastolic bp	Live	60–120; HTN ≥90; Severe ≥110	ACOG PB 222: severe DBP ≥110. Same link as above.

Column	Frequency	Criteria Used	Verified Source Link + Extracted Fact Used
severe headache	Live	0/1; higher prob when BP high	ACOG PB 222: severe features include symptoms (e.g., headache). Same link above.
vision changes	Live	0/1; higher prob with severe BP	ACOG PB 222: visual symptoms are severe features. Same link above.
upper abdominal pain	Live	0/1; higher prob with severe BP	ACOG PB 222: RUQ/epigastric pain indicates severity. Same link above.
swelling face hands	Live	0/1; mild correlation with PE risk	StatPearls PE symptom context. https://www.ncbi.nlm.nih.gov/books/NBK570611/
proteinuria dipstick	Periodic	0/trace/1+/2+/3+	StatPearls HTN in Pregnancy: dipstick proteinuria and ≥1+ in some contexts. https://www.ncbi.nlm.nih.gov/books/NBK430839/
fasting glucose	Live	70–140; target ¡95	ACOG Diabetes FAQ: fasting target ¡95 mg/dL. https://www.acog.org/womens-health/faqs/pregnancy-with-type-1-or-type-2-diabetes
postprandial 1hr glucose	Live	90–220; target ¡140	ACOG Diabetes FAQ: 1-hour post-meal target ¡140 mg/dL. Same link as above.
food log text	Live	meal templates; high-carb → spikes	ACOG diabetes management emphasizes post-meal control; food log supports context. Same ACOG link above.
post meal walk minutes	Live (optional)	0–30 minutes	StatPearls GDM: lifestyle/exercise as management. https://www.ncbi.nlm.nih.gov/books/NBK545196/
sleep hours	Live (optional)	4–10 hours	Supportive context feature; no strict guideline threshold applied.
weight kg	Periodic	40–120; often missing	StatPearls Macrosomia: obesity/weight gain linked to macrosomia. https://www.ncbi.nlm.nih.gov/books/NBK557577/
maternal age	Profile	18–45	Baseline maternal demographic input.
pre pregnancy bmi	Profile	16–45	StatPearls Macrosomia: maternal obesity is a risk factor. Same link above.
gdm diagnosed	Profile	0/1 based on glucose patterns	ACOG diabetes targets guide “uncontrolled” criteria. https://www.acog.org/womens-health/faqs/pregnancy-with-type-1-or-type-2-diabetes

Column	Frequency	Criteria Used	Verified Source Link + Extracted Fact Used
hba1c	Periodic	4.5–7.5; missing often	StatPearls GDM: diabetes monitoring indicator context. https://www.ncbi.nlm.nih.gov/books/NBK545196/
preeclampsia risk tier	Label	Low/Medium/High	Derived using ACOG severe-range BP criteria (160/110) and symptom/proteinuria relevance. ACOG PB 222 link above.
macrosomia risk tier	Label	Low/Medium/High	Derived using ACOG pregnancy glucose targets and BMI/weight macro-risk logic. ACOG diabetes FAQ + StatPearls Macrosomia.
preeclampsia risk score label	Label (optional)	0–1 (or 0–100)	Scaled using ACOG HTN thresholds (140/90, 160/110). ACOG PB 222 link above.
macrosomia risk score label	Label (optional)	0–1 (or 0–100)	Scaled using ACOG glucose targets (fasting ¡95, 1hr ¡140) + BMI risk. ACOG diabetes FAQ link above.

6 Conclusion

All 24 columns are fully documented with synthetic criteria and verified sources. The dataset intentionally prioritizes frequent live monitoring inputs (BP, glucose, symptoms, food logs) while allowing periodic updates for laboratory and manual inputs (proteinuria dipstick, HbA1c, weight). The clinical thresholds used for risk boundaries are taken from ACOG and NCBI StatPearls references to ensure validity and defendability.